

6.3 Solve the equations:

$$2x - y + z = 4$$

$$x + y + z = 3$$

$$3x - y - z = 1$$

```
Coefficients = [2 -1 1; 1 1 1; 3 -1 -1];  
Voltages = [4;3;1];  
X = Coefficients\Voltages;  
[x, y, z] = deal(X(1), X(2), X(3))
```

```
x = 1  
y = 0  
z = 2
```

Q:Write a function (find_prime_number) that finds if the input number to the function is a prime number or not. if the input number is a prime, it will display (P is a prime number) where P is the input number; otherwise; it will display (P is NOT a prime number).

```
prime(2)
```

```
this is a prime number
```

```
prime(3)
```

```
this is a prime number
```

```
prime(4)
```

```
this isn't a prime number
```

الدالة تكتب في ملف آخر ويتم استدعاؤها هنا وهذا كودها (يمكن استخدام الكود الآخر الموجود في حلول المد)

```
function prime(S)  
if isempty(find(primes(S) == S, 1))  
    disp("this isn't a prime number")  
else  
    disp("this is a prime number")  
end  
end
```

8.8 Write a program to compute the sum of the series $1^2 + 2^2 + 3^2 \dots$ such that the sum is as large as possible without exceeding 1000. The program should display how many terms are used in the sum.

```
n = 1^2;  
i = 2;  
while(n <= 1000)  
    n = n + i^2;  
    test = n + (i+1)^2;  
    if(test >= 1000)  
        disp(['The sum was ' num2str(n) ' with ' num2str(i) ' terms']);  
    end  
    i = i + 1;  
end
```

```

        break;
    end
    i = i + 1;
end

```

The sum was 819 with 13 terms

8.14 When a resistor (R), capacitor (C) and battery (V) are connected in series, a charge Q builds up on the capacitor according to the formula $Q(t) = CV (1 - e^{-t/RC})$ if there is no charge on the capacitor at time $t = 0$. The problem is to monitor the charge on the capacitor every 0.1 seconds in order to detect when it reaches a level of 8 units of charge, given that $V = 9$, $R = 4$, and $C = 1$. Write a program which displays the time and charge every 0.1 seconds until the charge first exceeds 8 units (i.e. the last charge displayed must exceed 8). Once you have done this, rewrite the program to display the charge only while it is strictly less than 8 units.

```

R = 4; V = 9; C = 1; t = 0; Q = 0;
while Q <= 8
    Q = C*V*(1-exp(-t/(R*C)));
    t = t + 0.1;
    Qtest = C*V*(1-exp(-t/(R*C)));
    if Qtest > 8
        break;
    end
end
disp(['Q (less than 8) = ' num2str(Q)]);

```

Q (less than 8) = 7.9775

```

disp(['Q (more than 8) = ' num2str(Qtest)]);

```

Q (more than 8) = 8.0028